



Programme: Data Science and Business Analytics

CALCULUS

Lecture 1 (07.09.2026)

**Introduction to Calculus.
Bounded sets. Real numbers**

1. Grades.

For the first half of the course (December)

$$G_1 = 0.2 * M_1 + 0.2 * Colloq_1 + 0.15 * Quiz_1 * (2 - Q_1) + 0.15 * HW_1 * Q_1 + 0.3 * E_1 + B_1.$$

with no rounding.

Final grade for semester 1 (formal):

$$Final_1 = \min \{Round(G_1), 10\}$$

with arithmetic rounding. Here

- M_1 is a grade obtained for the control work (midterm) after the end of the 1st module (from 0 to 10 each).
- $Colloq_1$ is a grade obtained for the colloquium at the end of November (from 0 to 10).
- $Quiz_1$ is the average of 70% your best grades (from 0 to 10) for regular weekly quizzes. Quizzes can be done during the seminar only and cannot be performed at any different time.
- $Q_1 = Quiz_1 / 10$
- E_1 is the exam grade (from 0 to 10) for the December exam.
- B_1 is the bonus grade (from 0 to 1) for the problems marked with $\star$ at each homework of the 1st semester and is equal to the part of all bonus problems passed.

For the second half of the course (June)

$$G_2 = 0.2 * M_2 + 0.2 * Colloq_2 + 0.15 * Quiz_2 * (2 - Q_1) + 0.15 * HW_2 * Q_2 + 0.3 * E_2 + B_2.$$

with no rounding.

Here

- M_2 is a grade obtained for the control work (midterm) after the end of the 3rd module (from 0 to 10 each).
- $Colloq_2$ is a grade obtained for the colloquium at the end of May (from 0 to 10).
- $Quiz_2$ is the average of 70% your best grades (from 0 to 10) for regular weekly quizzes. Quizzes can be done during the seminar only and cannot be performed at any different time.
- $Q_2 = Quiz_2 / 10$
- E_2 is the exam grade (from 0 to 10) for the June exam.
- B_2 is the bonus grade (from 0 to 1) for the problems marked with $\star$ at each homework of the 2nd semester and is equal to the part of all bonus problems passed.

Final course grade

$$Final = \text{round}\{0.4 * G_1 + 0.6 * G_2\}.$$

2. Bounded sets.

Let $\mathbb{R}$ be the set of all real numbers and $G \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$.

Definition 1. G is said to be *bounded above* if there is a number K such that $x \leq K$ for all $x \in G$. Using notations of $\forall$ (for any) and $\exists$ (there exists), we can write that G is *bounded above* if

$$\exists K \quad \forall x \in G : \quad x \leq K.$$

K is called an *upper bound* of G .

Definition 2. The set G is *bounded below* if there is a number L such that $x \geq L$ for $x \in G$:

$$\exists L \quad \forall x \in G : \quad x \geq L.$$

Such a number L is called a *lower bound* of G .

If G is bounded above and below, then G is *bounded*. In other words,

Definition 3. G is *bounded* if and only if there is $M > 0$ such that $|x| \leq M$ for all $x \in G$:

$$\exists M > 0 \quad \forall x \in G : \quad |x| \leq M.$$

A set G is called *unbounded* if it is not bounded (such M does not exist). In other words, G is called *unbounded* if

$$\forall M > 0 \quad \exists x \in G : \quad |x| > M.$$

Definition 4. Let G be a set bounded above. A number M is called the *least upper bound* or the *supremum* (*sup*) of G if both conditions hold

1. $x \leq M$ for all $x \in G$;
2. For any $\varepsilon > 0$ there is $x' \in G$: $x' > M - \varepsilon$.

We write the supremum of a set in the form: $M = \sup_{x \in G} x = \sup G$.

Definition 5. Let G be a set bounded below. A number N is called the *greatest lower bound* or the *infimum* (*inf*) of G if the following two conditions hold

1. $x \geq N$ for all $x \in G$.
2. For any $\varepsilon > 0$ there is $x' \in G$: $x' < N + \varepsilon$.

We write the infimum in the form: $N = \inf_{x \in G} x = \inf G$.

We assume that every nonempty subset of real numbers bounded above has a supremum and every nonempty subset of real numbers bounded below has a infimum.

References

- [1] Friedman A. Advanced Calculus. Dover Publications, INC, 1999 (F)
- [2] Stewart J. Calculus. Early Transcendentals. 6th edition. Thomson Brooks/Cole, 2008 (S)